

Context-induced belief geometry in LLMs

A project proposal

Xavier Poncini Dec 2, 2025

Computational mechanics (comp-mech) is a subfield of physics that studies convergent structures facilitating optimal prediction. Recent progress has demonstrated the relevance of comp-mech in describing the internal representations of toy-scale transformers predicting data from hidden Markov models (HMMs). In this work, it was demonstrated that neural networks represent 'belief state geometry' in their activations -- a well known structure in comp-mech, encoding distributions over hidden world states.



While these early indications are promising, it remains unclear whether production-scale transformers represent belief states in their activations. In this project, we seek to be the first to demonstrate the relevance of comp-mech to production-scale transformer activations. We will consider open-source LLMs, and investigate whether the most light-weight intervention -- prompting the model -- can be used to induce belief state geometry.

Overview. The core insight from comp-mech identified in [Shai et al.](#) can be summarised succinctly as:

For neural networks trained to near-optimal performance on HMM data, the belief state is linearly decodable from network activations.

In order for this insight to be valuable to the broader program of AI safety, it must be applicable to production-scale transformers. To evaluate whether LLMs represent belief geometry in their activations, we highlight three candidate approaches (in increasing order of intervention-complexity):

1. **Prompting.** Fix an HMM whose output token sequence is ergodic and an open-source LLM that has not been 'instruction fine-tuned'. Populate the context of this LLM with a sequence generated by the given HMM and evaluate the model-generated completion

with respect to the ground-truth distribution. If the model performs well at predicting the HMM data, use linear probes to identify belief state geometry.

2. **Natural tasks.** Identify tasks that are natural (in the [Anthropic sense](#)) to transformers, that also admit a formalisation in terms of an HMM e.g., modular addition. Prompt an LLM on this task, and use linear probes to identify the corresponding belief state geometry.
3. **Fine-tuning.** Fix an HMM and an open-source LLM. Fine-tune the LLM on the HMM data. Evaluate how this fine-tuning affects down-stream performance of the model on standard downstream tasks – if performance is not significantly altered, proceed to identifying belief state geometry in activations via linear probes.

In this project, we opt for the Nanda-style approach of [method minimalism](#) by attempting to identify belief state geometry using the simplest methods, then increasing as necessary.

Why prompting might just work. LLMs are capable of amazing things in-context, from learning [languages](#) to performing [linear regression](#). We are particularly inspired by recent results from [Park et al.](#) on the in-context learning of representations. In this work, LLMs are prompted with a sequence of data from a random walk on a fixed graph whose nodes are labelled by tokens with a distinct semantic meaning e.g., cat, ball, etc. As the model observes more context, the activations undergo a sudden reorganisation associated with the model discarding the semantic representation of the node indices, in favour of the new graph structure encoded by the input sequence.

This work demonstrates the remarkable ability of LLMs to restructure their internal representations in service of predicting tokens in context. For our use case, there are ways in which the restructuring of representations is both easier and harder than that explored by [Park et al.](#):

- **Easier.** We will use generic tokens for the vocabulary of the input sequence e.g. {0, 1, 2,...}, so the model will not have to significantly override the input semantics.
- **Harder.** We will consider HMM processes that require representations to track structures more elaborate than square and ring graphs.

Project milestones. Below we identify three key milestones that will guide the progression of the project.

1. **ICLR replication.** Replicate the findings of [Park et al.](#) on a suite of open-source LLMs but with graph nodes assigned generic labels e.g. {A, B, C, ...}.
2. **HMM identification.** Identify HMMs whose token outputs can be predicted well by LLMs in context, using [Dai et al.](#) as inspiration.
3. **Belief state probing.** Train linear probes to identify belief states in the activations of an LLM predicting HMM data.